

Lab Plan 7: Interactive Flutter UIs (Gestures & Sliders)

Course: Mobile Application Development

Duration: 3 Hours (180 Minutes)

Phase 1: GestureDetector

- **Task 1.1:** Create a Container (200x200) with a unique background color.
- **Task 1.2:** Wrap the container in a GestureDetector.
- **Task 1.3:** Logic Implementation:
 - **onTap:** Change the container's color to a random Colors value.
 - **onDoubleTap:** Toggle the borderRadius between 0 and 100 (Square to Circle).
 - **onLongPress:** Show a SnackBar saying "Resetting..." and return the box to its original state.

Phase 2: Sliders

- **Task 2.1:** Create a vertical Column containing a Text widget displaying a "Value" variable.
- **Task 2.2:** Add a Slider widget:
 - Set min to 0.0 and max to 100.0.
 - Set divisions to 10 to create discrete steps.
 - Update the "Value" text in real-time within onChanged.
- **Task 2.3:** Implement a CupertinoSlider below the Material one to observe the difference in native "look and feel."

Phase 3: The "Mood & Color" Mixer

The Challenge: Build a mini-app on a single screen where:

1. Three Sliders control the **Red**, **Green**, and **Blue** channels of a preview box.
2. A GestureDetector on the preview box allows the user to **Long Press** to "Copy" the Hex code to the console.
3. **Horizontal Drag (onHorizontalDragUpdate):** Should scale the size of the preview box up or down.